15 $f(n) = (2n + 1)^2 - (2n - 1)^2 - 10$ where n is an integer.

Prove that $f(n)$ is **never** a multiple of 8

	$2n$	1
$2n$	$4n^2$	$2n$
1	$2n$	1

	$2n$	-1
$2n$	$4n^2$	$-2n$
-1	$-2n$	1

$(2n + 1)^2 = 4n^2 + 4n + 1$

1 mark
Expand squares including middle terms.

$(2n - 1)^2 = 4n^2 - 4n + 1$

$f(n) = 4n^2 + 4n + 1 - (4n^2 - 4n + 1) - 10$

1 mark
Subtract every term in bracket.

$= 4n^2 + 4n + 1 - 4n^2 + 4n - 1 - 10$

$= 8n - 10 = 8(n - 2) + 6$

1 mark
Simplify fully.

$n - 2$ is an integer, so division by 8 leaves remainder 6.

Therefore $f(n)$ is never a multiple of 8.

1 mark
Non-zero remainder proves the claim.

(Total for Question 15 is 4 marks)